

IE507: Modeling and Computation Lab

Data Clustering and Visualization

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Question 1: Clustering using K-Means

1. Assumptions

- The dataset consists of continuous 2D spatial features where Euclidean distance is a valid similarity metric for grouping.
- The `k-means++` algorithm assumes that spreading out the initial cluster centers probabilistically will lead to a faster and more optimal convergence than purely random initialization.

2 & 3. Implementation and Plotting

- **Procedure:** The `scikit-learn` `KMeans` module was implemented to test the cluster ranges $K \in \{6, 7, 8, 9, 10, 11, 12, 13\}$. The `k-means++` initializer was utilized with a fixed random state for reproducibility.
- **Visualization:** A 2x4 grid of scatter plots was generated via `matplotlib`. Data points were color-coded based on their assigned cluster labels, and the final optimized centroids were superimposed using distinct red 'X' markers.

4. Observations on Increasing K (6 to 13) As the value of K increases from 6 to 13, the spatial segmentation of the data becomes progressively finer. At lower values (e.g., $K = 6$), the K-Means algorithm forces multiple distinct point clouds to merge into single, large clusters, resulting in high intra-cluster variance. As K increases toward 13, the centroids begin to align more accurately with the true local densities of the dataset. This reduces the Within-Cluster Sum of Squares (WCSS) and isolates individual groupings more effectively, though the data still appears slightly under-segmented at $K = 13$.

7. Procedure to Determine the Best Choice of Clusters

- **The Elbow Method:** Run the K-Means algorithm for an extended range of K values (e.g., $K = 2$ to 20). Calculate the WCSS (inertia) for each trained model. Plot the WCSS against K . The optimal number of clusters is typically located at the "elbow" or inflection point of the curve, where adding another cluster yields diminishing returns in variance reduction.
- **Silhouette Analysis:** Calculate the Silhouette Score for each K , which measures how similar an object is to its own cluster compared to other clusters. The K parameter that yields the highest average Silhouette Score represents the most mathematically optimal choice.

Question 2: Data Visualization - Cleveland Heart Data

1. DataFrame Dimensions: After loading the dataset, the pandas dataframe contains 303 rows and 14 columns, representing the individual patient records and their corresponding clinical features.

2. Object Type Columns: The columns `num_major_vessels`, `fluroscopy`, and `thal` are categorized as *object* types rather than numerical types because they contain missing values encoded as string characters (specifically the '?' symbol). Pandas defaults the entire column to an object type to accommodate these non-integer characters.

3 & 4. Age Histogram Observations: Using the default visualization settings, the bin size adjusts automatically. When forced to 50 bins, the bin width becomes much narrower (approximately 1 year per bin), exposing micro-fluctuations in the patient age distribution. The majority of the cohort falls between 50 and 60 years of age.

5. Purpose of KDE (Kernel Density Estimate): The KDE option smooths the discrete histogram bins into a continuous probability density curve. It is highly useful for visualizing the underlying shape and modality of the distribution, making it easier to identify skewness or multiple peaks without being distracted by arbitrary bin boundary choices.

9. Mean vs. Median Estimator in Bar Plots: Using a median estimator instead of the default mean estimator changes the bar plot heights because the median is robust to outliers. If the serum cholesterol data contains extreme high values, the mean bar plot will be artificially inflated, whereas the median bar plot provides a more accurate representation of the typical patient's cholesterol level.

14 & 15. Age vs. Serum Cholesterol Scatter Plot: The scatter plot reveals a weak positive correlation between age and serum cholesterol. The dark central line represents the linear regression fit (the line of best fit minimizing the residual sum of squares), while the light-colored bands surrounding it indicate the 95% confidence interval of that regression estimate.

16. Box Plot Interpretation: In the box plots depicting chest pain type against serum cholesterol:

- **Upper/Lower Boundaries:** Represent the 75th (Q3) and 25th (Q1) percentiles, forming the Interquartile Range (IQR).
- **Central Line:** Represents the median (50th percentile) value of the subset.
- **Points Beyond Error Bars:** These are statistically defined outliers, typically falling beyond $1.5 \times \text{IQR}$ from the upper or lower quartile hinges.

18. Violin Plot Observations: The violin plot effectively combines a standard box plot with a rotated kernel density plot on each side. When split by gender, it reveals that the density of cholesterol distributions varies significantly between male and female cohorts for specific chest pain types, highlighting wider variance and distinct probability peaks that traditional box plots hide.